

Dynamical Critical Phenomena

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Dynamic response and correlation functions

In order to describe critical dynamics, we first promote the field to be time-dependent, then we define two central quantities:

- 1 two-point two-time *correlation function* $C(\vec{x}, t; \vec{x}', t')$,
- 2 two-point two-time *response function* $G(\vec{x}, t; \vec{x}', t')$.

Relevant observables are written in terms of them (or higher-order ones).

At equilibrium, for a translation-invariant system, *Ergodic theorem* states

$$C(\vec{x} - \vec{x}', \tau) = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T \phi(\vec{x}, t' + \tau) \phi(\vec{x}', t') dt'. \quad (1)$$

which implies

$$C(\vec{x} - \vec{x}'; t - t') = \langle \phi(\vec{x}, t) \phi(\vec{x}', t') \rangle_{\text{eq}}, \quad (2)$$

Dynamic response and correlation functions

Adding a small time-dependent perturbation to the equilibrium system

$$S[\phi] \rightarrow S[\phi] - \int d\vec{x} h(\vec{x}, t) \phi(\vec{x}, t) \quad (3)$$

in the *linear response* regime, we obtain the response

$$\langle \phi(\vec{x}, t) \rangle = \int d\vec{x}' dt' G(\vec{x} - \vec{x}', t - t') h(\vec{x}', t') \quad (4)$$

which implies the definition of the *response function*

$$G(\vec{x} - \vec{x}', t - t') = \left. \frac{\delta \langle \phi(\vec{x}, t) \rangle}{\delta h(\vec{x}', t')} \right|_{h=0}. \quad (5)$$

REMARK: Because of causality, $G(\vec{x} - \vec{x}', t - t') = 0$ if $t < t'$.

Dynamic response and correlation functions

In equilibrium, the mechanism that generates fluctuations is the same governing dissipation (e.g. in Brownian motion velocity fluctuations are generated by collisions).

Fluctuation-dissipation relation:

$$\hat{C}(\vec{k}, \omega) = \frac{2k_B T}{\omega} \text{Im} \left[\hat{G}(\vec{k}, \omega) \right] \quad (6)$$

where

$$\hat{C}(\vec{k}, \omega) = \int_{-\infty}^{+\infty} d\tau \int d\vec{r} e^{i\omega\tau - i\vec{k}\cdot\vec{r}} C(\vec{r}, \tau) \quad (7)$$

$$\hat{G}(\vec{k}, \omega) = \int_{-\infty}^{+\infty} d\tau \int d\vec{r} e^{i\omega\tau - i\vec{k}\cdot\vec{r}} G(\vec{r}, \tau). \quad (8)$$

Dynamic response and correlation functions

REMARK: (A simple derivation of FDR)

Consider a single time-dependent Ising spin $s(t) \in \{-1, 1\}$.

If a small field h is switched on at $t = -\infty$ and off at $t = t'$, the system at time t' should be in equilibrium with

$$\Pr[s(t') = \pm 1] = \frac{e^{\pm\beta h}}{2 \cosh \beta h} \approx \frac{1}{2} (1 \pm \beta h + O(h^2)) \quad h \rightarrow 0 \quad (9)$$

So we have the average

$$\langle s(t) \rangle = \sum_{s(t')=\pm 1} \langle s(t) \rangle_{|s(t')} \Pr[s(t')] \quad (10)$$

$$= \frac{1}{2}(1 + \beta h) \langle s(t) \rangle_{|s(t')=+1} + \frac{1}{2}(1 - \beta h) \langle s(t) \rangle_{|s(t')=-1} \quad (11)$$

$$= \beta h \left[\frac{1}{2} \langle s(t) \rangle_{|s(t')=+1} - \frac{1}{2} \langle s(t) \rangle_{|s(t')=-1} \right] \quad (12)$$

$$= \beta h \langle s(t)s(t') \rangle_{\text{eq}} = \beta h C(t - t'). \quad (13)$$

Dynamic response and correlation functions

But we also have

$$\langle s(t) \rangle = \int_{-\infty}^t dt'' G(t - t'') h(t'') \quad (14)$$

$$= h \int_{-\infty}^t dt'' G(t - t'') \theta(t' - t'') \quad (15)$$

$$= h \int_{-\infty}^{t'} dt'' G(t - t''). \quad (16)$$

Combining the two,

$$C(t - t') = k_B T \int_{-\infty}^{t'} G(t - t'') dt'', \quad t > t'. \quad (17)$$

Since $C(t - t')$ is symmetric, setting $\tau = t - t'$, then

$$G(\tau) = -\beta \frac{dC(\tau)}{d\tau} \theta(\tau), \quad (18)$$

with $\theta(\tau) = 1$ if $\tau > 0$ and 0 otherwise.

Dynamic response and correlation functions

Fourier transforming,

$$\hat{G}(\omega) = -\beta \int_0^{+\infty} d\tau e^{i\omega\tau} \frac{dC(\tau)}{d\tau} \quad (19)$$

$$= -\beta C(0) + \beta \int_0^{+\infty} d\tau (i\omega) e^{i\omega\tau} C(\tau). \quad (20)$$

Finally we compute

$$\begin{aligned} \text{Im} [\hat{G}(\omega)] &= \frac{1}{2i} [\hat{G}(\omega) - \hat{G}^*(\omega)] \\ &= \frac{\beta}{2i} \left[\int_0^{+\infty} d\tau (i\omega) e^{i\omega\tau} C(\tau) - \int_0^{+\infty} d\tau (-i\omega) e^{-i\omega\tau} C(\tau) \right] \\ &= \frac{\beta}{2} \omega \int_{-\infty}^{+\infty} d\tau e^{i\omega\tau} C(\tau) \\ &= \frac{\beta}{2} \omega \hat{C}(\omega) \end{aligned}$$

which the FDR for a homogeneous system.

Phenomenological model of dissipative dynamics

Langevin equation for the velocity of a Brownian particle:

$$m \frac{d}{dt} v(t) = F(t) - \gamma v(t) + \eta(t), \quad (21)$$

with $\langle \eta(t) \rangle = 0$ and

$$\langle \eta(t) \eta(t') \rangle = \int D\eta \eta(t) \eta(t') e^{-\frac{1}{4k_B T \gamma} \int d\tau \eta(\tau)^2} = 2k_B T \gamma \delta(t - t'). \quad (22)$$

In the *overdamped limit*, i.e. $\tau^* = m/\gamma \ll 1$, implies

$$\frac{d}{dt} x(t) = \gamma^{-1} F(t) + \sqrt{2k_B T / \gamma} \eta(t), \quad (23)$$

with the Einstein relation $D = k_B T / \gamma$.

REMARK: Dissipation means $-\gamma v = -\gamma \frac{\partial}{\partial v} \left(\frac{1}{2} v^2 \right) = -\gamma \frac{\partial}{\partial v} E_{\text{kin}}$ not completely compensated by $F(t) = -\nabla V(x)$.

Phenomenological model of dissipative dynamics

Time-dependent generalization of the Landau approach:

- 1 a system evolving in contact with a thermal bath should tend to the equilibrium Gibbs distribution,
- 2 any conservation law present in the microscopic dynamics should be present at the coarse-grained level.

Consider the Landau-Ginzburg effective model

$$S[\varphi] = \beta \mathcal{H}_{\text{eff}}[\varphi] = \int d\vec{x} \left[\frac{1}{2} (\nabla \varphi)^2 + V[\varphi] \right], \quad (24)$$

with “potential”

$$V[\varphi(\vec{x})] = \frac{\mu^2}{2} \varphi(\vec{x})^2 + \frac{\lambda}{4!} \varphi(\vec{x})^4 - J(\vec{x}, t) \varphi(\vec{x}, t) \quad (25)$$

where $\mu^2 = \mu^2(T) \propto (T - T_0)$ [also $\mu^2 = rt/c$, $\lambda = 3!u/c^2$, $\varphi = c^{1/2}\phi$].

Phenomenological model of dissipative dynamics

The *time-dependent Landau-Ginzburg (TDLG) model* is described by the Langevin-type equation,

$$\frac{\partial}{\partial t} \varphi(\vec{x}, t) = - \int \Gamma(\vec{x} - \vec{x}') \frac{\delta \mathcal{H}_{\text{eff}}[\varphi]}{\delta \varphi(\vec{x}', t)} d\vec{x}' + \eta(\vec{x}, t) \quad (26)$$

$$= - \int D(\vec{x} - \vec{x}') \frac{\delta}{\delta \varphi(\vec{x}', t)} \left[\int d\vec{y} \left[\frac{1}{2} (\nabla \varphi(\vec{y}))^2 + V(\varphi(\vec{y})) \right] \right] d\vec{x}' + \eta(\vec{x}, t) \quad (27)$$

with *kinetic coefficient* Γ and space-dependent Gaussian noise

$$\begin{aligned} \langle \eta(\vec{x}, t) \rangle &= 0 \\ \langle \eta(\vec{x}, t) \eta(\vec{x}', t') \rangle &= 2D \delta(\vec{x} - \vec{x}') \delta(t - t') \end{aligned}$$

Phenomenological model of dissipative dynamics

The last one can be derived as follows

$$\begin{aligned}\varphi(\vec{x}, t + \delta t) &= \varphi(\vec{x}, t) - \int_t^{t+\delta t} dt' \int \Gamma(\vec{x} - \vec{x}'') \frac{\delta \mathcal{H}_{\text{eff}}}{\delta \varphi(\vec{x}'', t')} d\vec{x}'' + \int_t^{t+\delta t} \eta(\vec{x}, t') dt' \\ &\approx \varphi(\vec{x}, t) - \delta t \int \Gamma(\vec{x} - \vec{x}'') \frac{\delta \mathcal{H}_{\text{eff}}}{\delta \varphi(\vec{x}'', t)} d\vec{x}'' + \int_t^{t+\delta t} \eta(\vec{x}, t') dt',\end{aligned}\quad (28)$$

which implies

$$\begin{aligned}&\langle \varphi(\vec{x}, t + \delta t) \varphi(\vec{x}', t + \delta t) \rangle - \langle \varphi(\vec{x}, t) \varphi(\vec{x}', t) \rangle \\ &\approx - \left\langle \varphi(\vec{x}, t) \int \Gamma(\vec{x}' - \vec{x}'') \frac{\delta \mathcal{H}_{\text{eff}}}{\delta \varphi(\vec{x}'', t)} d\vec{x}'' \right\rangle \delta t \\ &\quad - \left\langle \varphi(\vec{x}', t) \int \Gamma(\vec{x} - \vec{x}'') \frac{\delta \mathcal{H}_{\text{eff}}}{\delta \varphi(\vec{x}'', t)} d\vec{x}'' \right\rangle \delta t \\ &\quad + \int_t^{t+\delta t} dt' \int_t^{t+\delta t} dt'' \langle \eta(\vec{x}, t') \eta(\vec{x}', t'') \rangle,\end{aligned}\quad (29)$$

Phenomenological model of dissipative dynamics

Since noise is delta-correlated in time

$$\int_t^{t+\delta t} dt' \int_t^{t+\delta t} dt'' \langle \eta(\vec{x}, t') \eta(\vec{x}', t'') \rangle = \int_t^{t+\delta t} dt' [\langle \eta(\vec{x}, t') \eta(\vec{x}', t) \rangle \delta t] = \langle \eta(\vec{x}, t) \eta(\vec{x}', t) \rangle \delta t. \quad (30)$$

In equilibrium the l.h.s. vanishes, then by symmetry $\Gamma(\vec{x} - \vec{x}') = \Gamma(\vec{x}' - \vec{x})$, and using equipartition formula

$$2 \int \Gamma(\vec{x} - \vec{x}'') \left\langle \varphi(\vec{x}', t) \frac{\delta \mathcal{H}_{\text{eff}}}{\delta \varphi(\vec{x}'', t)} \right\rangle d\vec{x}'' = 2k_B T \int \Gamma(\vec{x} - \vec{x}'') \delta(\vec{x}' - \vec{x}'') d\vec{x}'' \quad (31)$$

$$= 2\Gamma(\vec{x} - \vec{x}') k_B T. \quad (32)$$

We obtained the correlator of the noise,

$$\langle \eta(\vec{x}, t) \eta(\vec{x}', t') \rangle = 2k_B T \Gamma(\vec{x} - \vec{x}') \delta(t - t'). \quad (33)$$

Phenomenological model of dissipative dynamics

We can distinguish two dynamical models:

- *Model A* (no conservation of the order parameter), so $\Gamma(\vec{x} - \vec{x}') = \Gamma\delta(\vec{x} - \vec{x}')$, and

$$\frac{\partial}{\partial t}\varphi(\vec{x}, t) = -\Gamma \frac{\delta\mathcal{H}_{\text{eff}}}{\delta\varphi(\vec{x}, t)} + \eta(\vec{x}, t), \quad (34)$$

with $\langle\eta(\vec{x}, t)\rangle = 0$, $\langle\eta(\vec{x}, t)\eta(\vec{x}', t')\rangle = 2k_B T\Gamma\delta(\vec{x} - \vec{x}')\delta(t - t')$.

- *Model B* (local conservation of the order parameter), so $\Gamma(\vec{x} - \vec{x}') = -\Gamma\nabla^2\delta(\vec{x} - \vec{x}')$,

$$\frac{\partial}{\partial t}\varphi(\vec{x}, t) = \Gamma\nabla^2 \frac{\delta\mathcal{H}_{\text{eff}}}{\delta\varphi(\vec{x}, t)} + \eta(\vec{x}, t), \quad (35)$$

with $\langle\eta(\vec{x}, t)\rangle = 0$, $\langle\eta(\vec{x}, t)\eta(\vec{x}', t')\rangle = -2k_B T\Gamma\nabla^2\delta(\vec{x} - \vec{x}')\delta(t - t')$.

Gaussian model

The (dynamical) Gaussian model can be analyzed exactly. Setting $\lambda = 0$ in the TDLG model ,

$$\frac{\partial}{\partial t} \hat{\varphi}(\vec{k}, t) = -\hat{\Gamma}_k \frac{\delta \mathcal{H}_{\text{eff}}}{\delta \hat{\varphi}(-\vec{k}, t)} + \hat{\eta}(\vec{k}, t), \quad (36)$$

$$= -\hat{D}_k \left[(k^2 + \mu^2) \hat{\varphi}(\vec{k}, t) - \hat{J}(\vec{k}, t) \right] + \hat{\eta}(\vec{k}, t) \quad (37)$$

$$\text{with } \hat{D}_k = k_B T \hat{\Gamma}_k = \begin{cases} k_B T \Gamma = D & \text{model A} \\ k_B T \Gamma k^2 = D k^2 & \text{model B} \end{cases}$$

$$\begin{aligned} \langle \hat{\eta}(\vec{k}, t) \hat{\eta}(\vec{k}', t') \rangle &= \begin{cases} 2D \delta(t - t') \int d\vec{x} d\vec{x}' e^{-i\vec{x} \cdot (\vec{k} + \vec{k}')} & \text{Model A} \\ -2D \delta(t - t') \int d\vec{x} d\vec{x}' \nabla^2 e^{-i\vec{x} \cdot (\vec{k} + \vec{k}')} & \text{Model B} \end{cases} \\ &= 2\hat{D}_k (2\pi)^d \delta(t - t') \delta(\vec{k} + \vec{k}'). \end{aligned} \quad (38)$$

Fourier modes are decoupled,

$$\frac{\partial}{\partial t} \hat{\varphi}(\vec{k}, t) = -\hat{D}_k (k^2 + \mu^2) \hat{\varphi}(\vec{k}, t) + [\hat{D}_k \hat{J}(\vec{k}, t) + \hat{\eta}(\vec{k}, t)] \quad (39)$$

$$= -\tau_k^{-1} \hat{\varphi}(\vec{k}, t) + [\hat{D}_k \hat{J}(\vec{k}, t) + \hat{\eta}(\vec{k}, t)] \quad (40)$$

the formal solution is

$$\varphi(\vec{k}, t) = \varphi(\vec{k}, 0) e^{-t/\tau_k} + \int_0^t dt' e^{-(t-t')/\tau_k} [\hat{D}_k \hat{J}(\vec{k}, t') + \hat{\eta}(\vec{k}, t')] \quad (41)$$

with *relaxation time* $\tau_k = [\hat{D}_k (k^2 + \mu^2)]^{-1} = [\hat{D}_k (k^2 + \xi^{-2})]^{-1}$.

Notice that at criticality

$$\tau_k \propto \begin{cases} k^{-2} & \text{model A} \\ k^{-4} & \text{model B} \end{cases} \quad (42)$$

We have the scaling relation (the slowest mode scales like $k \sim \xi^{-1}$)

$$\tau_k \propto \frac{\xi^{2+a}}{(k\xi)^a \left((k\xi)^2 + 1 \right)} \quad (43)$$

$$= \xi^z f(k\xi) \quad (44)$$

with $z = 2$ for model A and $z = 4$ for model B.

Gaussian model

The correlation function

$$\langle \hat{\varphi}(\vec{k}, t) \hat{\varphi}(\vec{k}', t') \rangle = \int_0^t dt_1 \int_0^{t'} dt_2 e^{-(t-t_1)/\tau_k - (t'-t_2)/\tau_{k'}} \langle \eta(\vec{k}, t_1) \eta(\vec{k}', t_2) \rangle \quad (45)$$

$$= (2\pi)^d \delta(\vec{k} + \vec{k}') 2\hat{D}_k e^{-(t+t')/\tau_k} \int_0^{\min(t, t')} dt_1 e^{2t_1/\tau_k} \quad (46)$$

$$= (2\pi)^d \delta(\vec{k} + \vec{k}') \hat{D}_k \tau_k \left(e^{-|t-t'|/\tau_k} - e^{-(t+t')/\tau_k} \right) \quad (47)$$

In the limit $t, t' \rightarrow \infty$ with $t - t'$ finite,

$$\langle \hat{\varphi}(\vec{k}, t) \hat{\varphi}(\vec{k}', t') \rangle = (2\pi)^d \delta(\vec{k} + \vec{k}') \frac{1}{\xi^{-2} + k^2} e^{-|t-t'|/\tau_k}. \quad (48)$$

From the definition $\langle \hat{\varphi}(\vec{k}, \omega) \hat{\varphi}(\vec{k}', \omega') \rangle = (2\pi)^d \delta(\vec{k} + \vec{k}') \delta(\omega + \omega') \hat{C}_0(\vec{k}, \omega)$,

$$\hat{C}_0(\vec{k}, \omega) = \frac{2}{\hat{D}_k} \frac{1}{\frac{\omega^2}{D_k^2} + (\xi^{-2} + k^2)^2}. \quad (49)$$

$$\begin{aligned}
 \langle \hat{\varphi}(\vec{k}, \omega) \hat{\varphi}(\vec{k}', \omega') \rangle &= \int_{-\infty}^{+\infty} dt \int_{-\infty}^{+\infty} dt' e^{i\omega t + i\omega' t'} \langle \hat{\varphi}(\vec{k}, t) \hat{\varphi}(\vec{k}', t') \rangle \\
 &= \frac{(2\pi)^d \delta(\vec{k} + \vec{k}')}{\xi^{-2} + k^2} \int_{-\infty}^{+\infty} dt \int_{-\infty}^{+\infty} dt' e^{i\omega t + i\omega' t'} e^{-|t-t'|/\tau_k} \\
 &= \frac{(2\pi)^d \delta(\vec{k} + \vec{k}')}{\xi^{-2} + k^2} \int_{-\infty}^{+\infty} dt \int_{-\infty}^{+\infty} dt' e^{i\omega(t-t') + i(\omega' + \omega)t'} e^{-|t-t'|/\tau_k} \\
 &= \frac{(2\pi)^d \delta(\vec{k} + \vec{k}')}{\xi^{-2} + k^2} \int_{-\infty}^{+\infty} dt' e^{i(\omega' + \omega)t'} \int_{-\infty}^{+\infty} ds e^{i\omega s} e^{-|s|/\tau_k} \\
 &= \frac{(2\pi)^d \delta(\vec{k} + \vec{k}')}{\xi^{-2} + k^2} \int_{-\infty}^{+\infty} dt' e^{i(\omega' + \omega)t'} \frac{2/\tau_k}{\tau_k^{-2} + \omega^2} \\
 &= (2\pi)^d \delta(\vec{k} + \vec{k}') \delta(\omega' + \omega) \frac{2\hat{D}_k}{\hat{D}_k^2 [\xi^{-2} + k^2]^2 + \omega^2} \\
 &= (2\pi)^d \delta(\vec{k} + \vec{k}') \delta(\omega' + \omega) \frac{2/\hat{D}_k}{[\xi^{-2} + k^2]^2 + (\omega/\hat{D}_k)^2} \tag{50}
 \end{aligned}$$

Gaussian model

For the response function, we add an instantaneous perturbation

$$\hat{J}(\vec{k}, t) = \hat{J}(\vec{k})\delta(t) \quad (51)$$

Then,

$$\frac{\partial}{\partial t} \langle \hat{\varphi}(\vec{k}, t) \rangle = -\hat{D}_k (k^2 + \xi^{-2}) \langle \hat{\varphi}(\vec{k}, t) \rangle - \hat{D}_k \hat{J}(\vec{k}), \quad (52)$$

$$\implies -i\omega \langle \hat{\varphi}(\vec{k}, \omega) \rangle = -\hat{D}_k (k^2 + \xi^{-2}) \langle \hat{\varphi}(\vec{k}, \omega) \rangle - \hat{D}_k \hat{J}(\vec{k}) \quad (53)$$

with solution

$$\langle \hat{\varphi}(\vec{k}, \omega) \rangle = \frac{\hat{J}(\vec{k})}{-\frac{i\omega}{\hat{D}_k} + k^2 + \xi^{-2}}. \quad (54)$$

Gaussian model

From the definition of response function, $\langle \hat{\varphi}(\vec{k}, \omega) \rangle = \hat{G}_0(\vec{k}, \omega) \hat{J}(\vec{k})$, then

$$\hat{G}_0(\vec{k}, \omega) = \frac{1}{-\frac{i\omega}{\hat{D}_k} + k^2 + \xi^{-2}}. \quad (55)$$

In the static limit ($\omega = 0$), we recover Ornstein-Zernicke form of Gaussian propagator.

For $\xi \rightarrow \infty$, the response function can be put in the scaling form ($a = 0, 2$)

$$\hat{G}_0(\vec{k}, \omega) = \frac{\hat{D}_k}{-i\omega + \hat{D}_k k^2} = \frac{Dk^a}{-i\omega + Dk^{2+a}} \quad (56)$$

$$= \frac{k^{-2}D}{D - i\omega k^{-2}} \quad (57)$$

$$= k^{-2+\eta} \tilde{\Phi}(\omega k^{-2}), \quad \eta = 0. \quad (58)$$

From previous results we recover the *fluctuation-dissipation relation*,

$$\hat{C}_0(\vec{k}, \omega) = \frac{2}{\hat{D}_k} \frac{1}{\frac{\omega^2}{\hat{D}_k^2} + (\xi^{-2} + k^2)^2}. \quad (59)$$

$$= \frac{2}{\omega} \frac{\omega / \hat{D}_k}{\frac{\omega^2}{\hat{D}_k^2} + (\xi^{-2} + k^2)^2} \quad (60)$$

$$= \frac{2}{\omega} \text{Im} \left[\hat{G}_0(\vec{k}, \omega) \right]. \quad (61)$$

Dynamic scaling hypothesis

Following what we obtained for the Gaussian model, we can generalize the scaling hypothesis to include time dependence.

In the critical region, both correlation length ξ and relaxation time τ_k are finite

$$\hat{G}(\vec{k}, \omega, \xi) = \xi^{2-\eta} \Phi(\xi \vec{k}, \xi^z \omega) \quad (62)$$

with relaxation time $\tau_k = \xi^z f(k\xi)$.

Eliminating dependence on ξ choosing $k \sim \xi^{-1}$, we find the critical scaling

$$\hat{G}(\vec{k}, \omega) = k^{-2+\eta} \tilde{\Phi}(\omega k^{-z}). \quad (63)$$

Dynamic scaling hypothesis

Consider the Ising model in $d = 1$, relaxation implies energy dissipation by means of annihilation of domain walls.

Say $\tau_{\text{rel}} \sim \Gamma^{-1}$ is time scale of domain wall motion. If we coarse-grain ℓ spins together, the time is $\tau'_{\text{rel}} \sim \ell^2 \Gamma^{-1}$.

Hence, under RG, the kinetic coefficient transforms as follows

$$\Gamma \rightarrow \Gamma' = \Gamma \ell^{-z}, \quad z = 2.$$

In case of non-gaussian fluctuations,

$$\Gamma' = \Gamma \mathcal{F}[\ell, \{\mathcal{P}\}],$$

where $\mathcal{F}[\ell, \{\mathcal{P}\}]$ is a non-dimensional function of parameters and scale factor ℓ (not on Γ).

Perturbative approach for Model A

Write the full Langevin equation of Model A in the form

$$\left(\frac{1}{D} \frac{\partial}{\partial t} - \nabla^2 + \mu^2 \right) \varphi(\vec{x}, t) = -\frac{\lambda}{6} \varphi(\vec{x}, t)^3 + \frac{1}{D} \eta(\vec{x}, t) + J(\vec{x}, t), \quad (64)$$

then the formal solution can be expressed in terms of the Gaussian response function,

$$\varphi(\vec{x}, t) = \int d\vec{x}' \int_0^t dt' G_0(\vec{x} - \vec{x}', t - t') \left[-\frac{\lambda}{6} \varphi(\vec{x}', t')^3 + \frac{1}{D} \eta(\vec{x}', t') + J(\vec{x}', t') \right], \quad (65)$$

in which

$$G_0(\vec{x} - \vec{x}'; t - t') = \frac{D}{(4\pi)^{d/2}} e^{-\frac{(t-t')}{\xi^2} - \frac{(\vec{x}-\vec{x}')^2}{4D(t-t')}} (D(t-t'))^{-\frac{d}{2}}. \quad (66)$$

Perturbative approach for Model A

The previous equation in Fourier space becomes,

$$\begin{aligned} & \left[-\frac{i\omega}{D} + k^2 + \mu^2 \right] \hat{\varphi}(\vec{k}, \omega) \\ &= -\frac{\lambda}{6} \int \frac{d\vec{k}_1 d\omega_1}{(2\pi)^{d+1}} \frac{d\vec{k}_2 d\omega_2}{(2\pi)^{d+1}} \hat{\varphi}(\vec{k}_1, \omega_1) \hat{\varphi}(\vec{k}_2, \omega_2) \hat{\varphi}(\vec{k} - \vec{k}_1 - \vec{k}_2, \omega - \omega_1 - \omega_2) \\ & \quad + \frac{1}{D} \hat{\eta}(\vec{k}, \omega) + \hat{J}(\vec{k}, \omega) \end{aligned} \tag{67}$$

which implies

$$\begin{aligned} \hat{\varphi}(\vec{k}, \omega) &= -\frac{\lambda}{6} \hat{G}_0(\vec{k}, \omega) \int \frac{d\vec{k}_1 d\omega_1}{(2\pi)^{d+1}} \frac{d\vec{k}_2 d\omega_2}{(2\pi)^{d+1}} \hat{\varphi}(\vec{k}_1, \omega_1) \hat{\varphi}(\vec{k}_2, \omega_2) \hat{\varphi}(\vec{k} - \vec{k}_1 - \vec{k}_2, \omega - \omega_1 - \omega_2) \\ & \quad + \frac{1}{D} \hat{G}_0(\vec{k}, \omega) \hat{\eta}(\vec{k}, \omega) + \hat{G}_0(\vec{k}, \omega) \hat{J}(\vec{k}, \omega) \end{aligned} \tag{68}$$

Perturbative approach for Model A

The formal solution can be written diagrammatically using,

$$G_0(\vec{x} - \vec{x}', t - t') = \begin{array}{c} (x, t) \qquad (x', t') \\ \bullet \longleftarrow \bullet \end{array}$$

$$\hat{G}_0(\vec{k}, \omega) = \begin{array}{c} (\vec{k}, \omega) \\ \longrightarrow \longleftarrow \end{array}$$

$$\eta(\vec{x}, t) = \begin{array}{c} (\vec{x}, t) \\ \longrightarrow \longleftarrow \otimes \end{array}$$

$$\hat{\eta}(\vec{k}, \omega) = \begin{array}{c} (\vec{k}, \omega) \\ \longrightarrow \longleftarrow \otimes \end{array}$$

$$J(\vec{x}, t) = \begin{array}{c} (\vec{x}, t) \\ \longrightarrow \bullet \end{array}$$

$$\hat{J}(\vec{k}, \omega) = \begin{array}{c} (\vec{k}, \omega) \\ \longrightarrow \bullet \end{array}$$

Perturbative approach for Model A

$$\begin{aligned}\hat{\varphi}(\vec{k}, \omega) &= \hat{G}_0(\vec{k}, \omega) \hat{J}(\vec{k}, \omega) + \frac{1}{D} \hat{G}_0(\vec{k}, \omega) \hat{\eta}(\vec{k}, \omega) \\ &\quad - \frac{\lambda}{6} \hat{G}_0(\vec{k}, \omega) \int \frac{d\vec{k}_1 d\omega_1}{(2\pi)^{d+1}} \frac{d\vec{k}_2 d\omega_2}{(2\pi)^{d+1}} \hat{G}_0(\vec{k}_1, \omega_1) \hat{J}(\vec{k}_1, \omega_1) \hat{G}_0(\vec{k}_2, \omega_2) \hat{J}(\vec{k}_2, \omega_2) \\ &\quad \times \hat{G}_0(\vec{k} - \vec{k}_1 - \vec{k}_2, \omega - \omega_1 - \omega_2) \hat{J}(\vec{k} - \vec{k}_1 - \vec{k}_2, \omega - \omega_1 - \omega_2) + \dots\end{aligned}\quad (69)$$

$$\begin{aligned}\hat{\varphi}(\vec{k}, \omega) &= \text{diagram 1} + \frac{1}{D} \text{diagram 2} - \frac{\lambda}{6} \text{diagram 3} - 3 \frac{1}{D} \frac{\lambda}{6} \text{diagram 4} \\ &\quad - 3 \frac{1}{D^2} \frac{\lambda}{6} \text{diagram 5} - \frac{1}{D^3} \frac{\lambda}{6} \text{diagram 6} + \dots\end{aligned}$$

Perturbative approach for Model A

Loops appear when averaging over the noise (prefactor $\frac{1}{D} \times \frac{1}{D} \times 2D$)

The diagram shows a vertex (represented by a circle with a cross) with two outgoing lines. One line is solid and the other is dashed. This is equal to $\frac{2}{D}$ times a loop diagram (a circle with two arrows pointing in opposite directions).

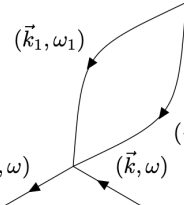
$$\begin{aligned}
 & \left\langle -\frac{1}{2} \frac{\lambda}{D^2} \int \frac{d\vec{k}_1 d\omega_1}{(2\pi)^{d+1}} \frac{d\vec{k}_2 d\omega_2}{(2\pi)^{d+1}} \hat{G}_0(\vec{k}_1, \omega_1) \hat{\eta}(\vec{k}_1, \omega_1) \hat{G}_0(\vec{k}_2, \omega_2) \hat{\eta}(\vec{k}_2, \omega_2) \right\rangle \\
 &= -\frac{1}{2} \frac{\lambda}{D^2} \int \frac{d\vec{k}_1 d\omega_1}{(2\pi)^{d+1}} \frac{d\vec{k}_2 d\omega_2}{(2\pi)^{d+1}} \hat{G}_0(\vec{k}_1, \omega_1) \hat{G}_0(\vec{k}_2, \omega_2) \left\langle \hat{\eta}(\vec{k}_1, \omega_1) \hat{\eta}(\vec{k}_2, \omega_2) \right\rangle \\
 &= -\frac{1}{2} \frac{\lambda}{D^2} 2D(2\pi)^{d+1} \int \frac{d\vec{k}_1 d\omega_1}{(2\pi)^{d+1}} \frac{d\vec{k}_2 d\omega_2}{(2\pi)^{d+1}} \hat{G}_0(\vec{k}_1, \omega_1) \hat{G}_0(\vec{k}_2, \omega_2) \delta(\vec{k}_1 + \vec{k}_2) \delta(\omega_1 + \omega_2) \\
 &= -\frac{1}{2} \frac{2\lambda}{D} \int \frac{d\vec{k}_1 d\omega_1}{(2\pi)^{d+1}} \hat{G}_0(\vec{k}_1, \omega_1) \hat{G}_0(-\vec{k}_1, -\omega_1)
 \end{aligned}$$

(70)

Perturbative approach for Model A

The response function $\hat{G}(\vec{k}, \omega)$ of the full theory can be also computed perturbatively.

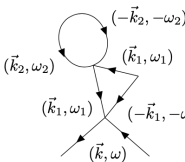
At order $O(\lambda)$ we have the correction


$$-\frac{\lambda}{D} (\vec{k}, \omega) (\vec{k}, \omega) = -\frac{\lambda}{D} \hat{G}_0(\vec{k}, \omega)^2 \int \frac{d\vec{k}_1 d\omega_1}{(2\pi)^{d+1}} \hat{G}_0(\vec{k}_1, \omega_1) \hat{G}_0(-\vec{k}_1, -\omega_1)$$

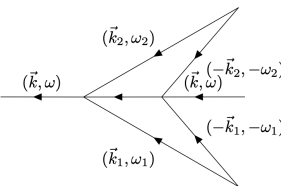
REMARK: In the static limit ($\omega = 0$), we recover the 1-loop correction to the static propagator.

Perturbative approach for Model A

At order $O(\lambda^2)$ we have two corrections:



$$18 \left(-\frac{\lambda}{3D} \right)^2 (\vec{k}, \omega) = \frac{\lambda^2}{D^2} \hat{G}_0(\vec{k}, \omega)^2 \int \frac{d\vec{k}_1 d\omega_1}{(2\pi)^{d+1}} \hat{G}_0(\vec{k}_1, \omega_1) \hat{G}_0(-\vec{k}_1, -\omega_1) \\ \times \left[\hat{G}_0(\vec{k}_1, \omega_1) + \hat{G}_0(-\vec{k}_1, -\omega_1) \right] \int \frac{d\vec{k}_2 d\omega_2}{(2\pi)^{d+1}} \hat{G}_0(\vec{k}_2, \omega_2) \hat{G}_0(-\vec{k}_2, -\omega_2).$$



$$\left(-\frac{\lambda}{3D} \right)^2 (\vec{k}, \omega) = \frac{\lambda^2}{9D^2} \hat{G}_0(\vec{k}, \omega)^2 \int \frac{d\vec{k}_1 d\omega_1}{(2\pi)^{d+1}} \frac{d\vec{k}_2 d\omega_2}{(2\pi)^{d+1}} \hat{G}_0(\vec{k}_1, \omega_1) \hat{G}_0(-\vec{k}_1, -\omega_1) \\ \times \hat{G}_0(\vec{k}_2, \omega_2) \hat{G}_0(-\vec{k}_2, -\omega_2) \hat{G}_0(\vec{k} - \vec{k}_1 - \vec{k}_2, \omega - \omega_1 - \omega_2)$$

Perturbative approach for Model A

Collecting the expression up to $O(\lambda^2)$ we find

$$\begin{aligned} G(\vec{k}, \omega) = & \hat{G}_0(\vec{k}, \omega) \left\{ 1 - \frac{\lambda}{D} \hat{G}_0(\vec{k}, \omega) \int \frac{d\vec{k}_1 d\omega_1}{(2\pi)^{d+1}} \hat{G}_0(\vec{k}_1, \omega_1) \hat{G}_0(-\vec{k}_1, -\omega_1) \right. \\ & + \frac{\lambda^2}{D^2} \hat{G}_0(\vec{k}, \omega) \int \frac{d\vec{k}_1 d\omega_1}{(2\pi)^{d+1}} \hat{G}_0(\vec{k}_1, \omega_1) \hat{G}_0(-\vec{k}_1, -\omega_1) [\hat{G}_0(\vec{k}_1, \omega_1) + \hat{G}_0(-\vec{k}_1, -\omega_1)] \\ & \int \frac{d\vec{k}_2 d\omega_2}{(2\pi)^{d+1}} \hat{G}_0(\vec{k}_2, \omega_2) \hat{G}_0(-\vec{k}_2, -\omega_2) \\ & + \frac{2\lambda^2}{3D^2} \hat{G}_0(\vec{k}, \omega)^2 \int \frac{d\vec{k}_1 d\omega_1}{(2\pi)^{d+1}} \frac{d\vec{k}_2 d\omega_2}{(2\pi)^{d+1}} \frac{d\vec{k}_3 d\omega_3}{(2\pi)^{d+1}} \delta\left(\sum_{i=1}^3 \vec{k}_i - \vec{k}\right) \delta\left(\sum_{i=1}^3 \omega_i - \omega\right) \\ & \left. \times [\hat{G}_0(\vec{k}_1, \omega_1) \hat{G}_0(-\vec{k}_1, -\omega_1) \hat{G}_0(\vec{k}_2, \omega_2) \hat{G}_0(-\vec{k}_2, -\omega_2) \hat{G}_0(\vec{k}_3, \omega_3) + 2 \text{ perms.}] \right\} \end{aligned} \quad (71)$$

With the same approach we can compute $C(\vec{k}, \omega)$ and verify that FDR is satisfied up to $O(\lambda^2)$.

Response Functional Formalism

Functional formalism introduced by Martin, Siggia and Rose (1973) to study critical behavior of fluids.

Formally, the correlation function should be defined as

$$C(\vec{x}_1, \vec{x}_2; t_1, t_2) = \left\langle \int D\varphi \delta[\varphi(\vec{x}, t) = \text{solution}] \varphi(\vec{x}_1, t_1) \varphi(\vec{x}_2, t_2) \right\rangle_{\eta} \quad (72)$$

but we do not know the explicit solution. A change of variables is needed.

So, we impose the stochastic equation itself,

$$1 = Z \equiv \left\langle \int D\varphi \delta[\varphi(\vec{x}, t) = \text{solution}] \right\rangle_{\eta} \quad (73)$$

$$\left\langle \int D\varphi \det \mathcal{J}[\varphi] \delta \left[\dot{\varphi} + D \frac{\delta \beta \mathcal{H}_{\text{eff}}}{\delta \varphi} - \eta \right] \right\rangle_{\eta} \quad (74)$$

Response Functional Formalism

It comes from properties of delta function: if $\vec{y} = \vec{h}(\vec{x})$, then

$$\int \delta(\vec{y}) F(\vec{y}) d\vec{y} = \int \delta(\vec{h}(\vec{x})) F(\vec{h}(\vec{x})) \left| \det \nabla \vec{h}(\vec{x}) \right| d\vec{x} \quad (75)$$

Consider the operator $\mathcal{N}[\varphi] = \dot{\varphi} + D \frac{\delta \beta \mathcal{H}_{\text{eff}}}{\delta \varphi} - \eta$ discretized according to forward differences ($\varphi_j = \varphi(t_j)$ and $\eta_j = \eta(t_j)$)

$$\mathcal{N}_j[\varphi] = \varphi_j - \varphi_{j-1} - \delta t \left[-D \frac{\delta \beta \mathcal{H}_{\text{eff}}}{\delta \varphi_{j-1}} + \eta_{j-1} \right]. \quad (76)$$

The Jacobian has elements

$$\mathcal{J}_{jl} = \frac{\partial \mathcal{N}_j}{\partial \varphi_l} = \begin{cases} 1 & \text{if } l = j \\ -1 + \delta t D \frac{\delta^2 \beta \mathcal{H}_{\text{eff}}}{\delta \varphi_{j-1}^2} & \text{if } l = j - 1 \\ 0 & \text{otherwise} \end{cases}$$

where and $\partial \eta_{j-1} / \partial \varphi_l = 0$ for $l \geq j - 1$ by causality (*Ito regularization*). The matrix is a lower triangular, so $\det \mathcal{J}[\varphi] = 1$.

Response Functional Formalism

Using the integral representation of delta function,

$$\delta \left[\dot{\varphi} + D \frac{\delta \beta \mathcal{H}_{\text{eff}}}{\delta \varphi} - \eta \right] = \int D\tilde{\varphi} \exp \left[- \int d\vec{x} dt \tilde{\varphi} \left(\dot{\varphi} + D \frac{\delta \beta \mathcal{H}_{\text{eff}}}{\delta \varphi} - \eta \right) \right] \quad (77)$$

we have now a functional integral over a conjugate field $\tilde{\varphi}(\vec{x}, t)$, also called *response field*.

Performing the average over the Gaussian noise,

$$Z = \left\langle \int D\varphi \int D\tilde{\varphi} e^{-\int d\vec{x} dt \tilde{\varphi} \left(\dot{\varphi} + D \frac{\delta \beta \mathcal{H}_{\text{eff}}}{\delta \varphi} - \eta \right)} \right\rangle_{\eta} \quad (78)$$

$$= \int D\eta \int D\varphi \int D\tilde{\varphi} e^{-\int d\vec{x} dt \left(\tilde{\varphi} \dot{\varphi} + D \tilde{\varphi} \frac{\delta \beta \mathcal{H}_{\text{eff}}}{\delta \varphi} \right) + \int d\vec{x} dt \left(\tilde{\varphi} \eta - \frac{\eta^2}{4D} \right)} \quad (79)$$

$$= \int D\varphi \int D\tilde{\varphi} e^{-\int d\vec{x} dt \left(\tilde{\varphi} \dot{\varphi} + D \tilde{\varphi} \frac{\delta \beta \mathcal{H}_{\text{eff}}}{\delta \varphi} \right) + \int d\vec{x} dt (D \tilde{\varphi}^2)} \quad (80)$$

Response Functional Formalism

We found a statistical measure over stochastic trajectories

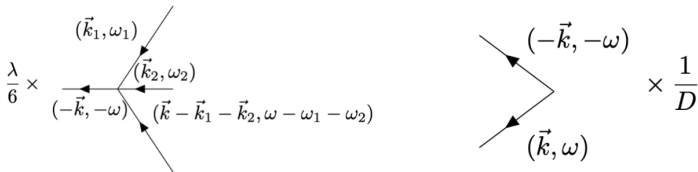
$$\int D\varphi \int D\tilde{\varphi} \exp[-S[\varphi, \tilde{\varphi}]] \quad (81)$$

with action

$$\tilde{S}[\varphi, \tilde{\varphi}] = \int d\vec{x} dt \left[\tilde{\varphi} \left(\frac{1}{D} \partial_t \varphi + \left(-\nabla^2 \varphi + \mu^2 \varphi + \frac{\lambda}{6} \varphi^3 - J \right) \right) - \frac{1}{D} \tilde{\varphi}^2 \right] \quad (82)$$

where we rescaled $\tilde{\varphi} \rightarrow \tilde{\varphi}/D$ for later convenience.

Vertices of the perturbation theory:



Response Functional Formalism

Dimensional analysis: given a momentum scale Λ ,

$$[\tilde{\varphi}\varphi] = [d^d x]^{-1} [dt]^{-1} \Lambda^{-2} = \Lambda^{d-2} [\text{time}]^{-1}$$

Time is a fundamental quantity, independent of momentum scale, then $[\tilde{\varphi}\varphi] = \Lambda^{d-2}$, which implies

$$[D] = \Lambda^{-2} \quad (83)$$

and $[\tilde{\varphi}^2][D]^{-1} = \Lambda^d [\text{time}]^{-1} = \Lambda^d$, that gives

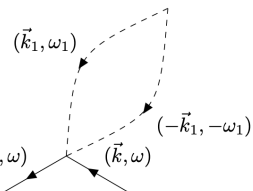
$$[\varphi] = \Lambda^{\frac{d}{2}-1}, \quad [\tilde{\varphi}] = \Lambda^{\frac{d}{2}-1} \quad \left[[\tilde{\varphi}] = \Lambda^{\frac{d}{2}+1} \quad \text{if not rescaled} \right] \quad (84)$$

The other parameters scale as we already know, $[\mu^2] = \Lambda^2$ and $[\lambda] = \Lambda^{4-d}$.

The upper critical dimension of the dynamic theory is $d = 4$, so we can apply perturbative RG for $d \leq 4$.

Response Functional Formalism and RG

Corrections to μ^2 at $O(\lambda)$ can be obtained as follows. Consider the diagram,



$$\frac{\lambda}{D} \text{ (diagram) } = \frac{\lambda}{D} \hat{G}_{0,<}(\vec{k}, \omega)^2 \int_{\Lambda/\ell}^{\Lambda} \frac{d\vec{k}_1}{(2\pi)^d} \int \frac{d\omega_1}{2\pi} \hat{G}_{0,>}(\vec{k}_1, \omega_1) \hat{G}_{0,>}(-\vec{k}_1, -\omega_1)$$

Then we have

$$G(\vec{k}, \omega) = \hat{G}_{0,<}(\vec{k}, \omega) \left\{ 1 - \frac{\lambda}{D} \hat{G}_{0,<}(\vec{k}, \omega) \int_{\Lambda/\ell}^{\Lambda} \frac{d\vec{k}_1}{(2\pi)^d} \int \frac{d\omega_1}{2\pi} \hat{G}_{0,>}(\vec{k}_1, \omega_1) \hat{G}_{0,>}(-\vec{k}_1, -\omega_1) \right\} \quad (85)$$

Response Functional Formalism and RG

$$\begin{aligned}
 G(\vec{k}, \omega)^{-1} &= \hat{G}_{0,<}(\vec{k}, \omega)^{-1} \left\{ 1 + \frac{\lambda}{D} \hat{G}_{0,<}(\vec{k}, \omega) \int_{\Lambda/\ell}^{\Lambda} \frac{d\vec{k}_1}{(2\pi)^d} \int \frac{d\omega_1}{2\pi} \hat{G}_{0,>}(\vec{k}_1, \omega_1) \hat{G}_{0,>}(-\vec{k}_1, -\omega_1) \right\} \\
 &= \hat{G}_{0,<}(\vec{k}, \omega)^{-1} + \frac{\lambda}{D} \int_{\Lambda/\ell}^{\Lambda} \frac{d\vec{k}_1}{(2\pi)^d} \int \frac{d\omega_1}{2\pi} \hat{G}_{0,>}(\vec{k}_1, \omega_1) \hat{G}_{0,>}(-\vec{k}_1, -\omega_1) + O(\lambda^2) \\
 -\frac{i\omega}{D_1} + k^2 + \mu_1^2 &= -\frac{i\omega}{D} + k^2 + \mu^2 + \frac{\lambda}{D} \int_{\Lambda/\ell}^{\Lambda} \frac{d\vec{k}_1}{(2\pi)^d} \int \frac{d\omega_1}{2\pi} \frac{1}{-\frac{i\omega_1}{D} + k_1^2 + \mu^2} \frac{1}{\frac{i\omega_1}{D} + k_1^2 + \mu^2} + O(\lambda^2) \\
 &= -\frac{i\omega}{D} + k^2 + \mu^2 + \frac{\lambda}{D} \int_{\Lambda/\ell}^{\Lambda} \frac{d\vec{k}_1}{(2\pi)^d} \int \frac{d\omega_1}{2\pi} \frac{1}{\left(\frac{\omega_1}{D}\right)^2 + (k_1^2 + \mu^2)^2} + O(\lambda^2) \\
 &= -\frac{i\omega}{D} + k^2 + \mu^2 + \frac{\lambda}{2D} \int_{\Lambda/\ell}^{\Lambda} \frac{d\vec{k}_1}{(2\pi)^d} \frac{1}{k_1^2 + \mu^2} + O(\lambda^2)
 \end{aligned}$$

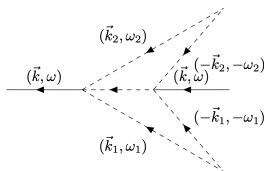
so at $O(\lambda)$, we have $D_1 = D$ and

$$\mu_1^2 = \mu^2 + \frac{\lambda}{2D} \int_{\Lambda/\ell}^{\Lambda} \frac{d\vec{k}_1}{(2\pi)^d} \frac{1}{k_1^2 + \mu^2} \quad (86)$$

which is the already known shift in the critical temperature.

Response Functional Formalism and RG

The kinetic coefficient D gets affected at $O(\lambda^2)$, from the diagram



Then we compute the coarse-grained parameter from

$$\frac{1}{D_1} = \frac{\partial}{\partial(-i\omega)} \left[\hat{G}(\vec{k}, \omega) \right]^{-1} \Big|_{\omega=0, k^2=0} \quad (87)$$

and after a tedious calculation, from the renormalization of D , we get the *dynamical critical exponent for model A* in $d = 4 - \epsilon$

$$z = 2 + \eta \left(-1 + 6 \log \frac{4}{3} \right) = 2 + \frac{\epsilon^2}{54} \left(-1 + 6 \log \frac{4}{3} \right) \approx 2.0135 \quad (88)$$

Response Functional Formalism and RG

For Model B, since the Gaussian propagator is

$$\hat{G}_0(\vec{k}, \omega) = \frac{1}{-i\omega/D + k^2(k^2 + \mu^2)}, \quad (89)$$

and all vertices are proportional to k^2 (interaction $-\lambda k^2$, noise $2k^2/D$), all perturbative corrections vanish for $k^2 \rightarrow 0$.

There is no correction to the kinetic coefficient D under RG, only usual field renormalization at second order.

The *dynamic critical exponent for model B* is then, at all orders in perturbation,

$$z = 4 - \eta. \quad (90)$$